A Note on This Document

This package is written for two kinds of readers whose roles, though distinct, are inseparable in any deployment of this pathway.

Type D readers hold the position to decide for a system to follow a path. They see a destination — farmers reached, institutions aligned, infrastructure built — and ask: wouldn't it be possible? They align institutional priorities and budgets. They hold a clear-eyed understanding of constraints and future possibilities. They inspire and declare.

Type C readers answer that question with the first level of the how. They map the terrain between here and there: what is feasible, what is viable, what the options are, and what those options ask of the institution. They produce the analysis that gives Type D's vision its foundations.

Refer to 4 types of readers below.

Both kinds of reader are present in every deployment this pathway documents. Sometimes they are the same person. Often they are not. What they share is the question that precedes everything else: is this real, is it feasible, and what would it take here?

This package presents what others have found when they asked those questions — and what followed from the answers.

What This Document Is — and Is Not

This is not a technical specification. It does not instruct construction.

This is not a case study. Case studies anchor to conclusions. This document opens — it presents what has been decided, what was found feasible, what was found hard, and what emerged. The reader draws their own conclusions.

This is not a document of advocacy. It does not argue that this pathway should be followed. The decision at every level belongs to the reader.

What this document is: a structured account of a pathway that others have walked — offered to inform those who hold the position to decide whether to walk it, and those who hold the responsibility to assess how.

The Constitution actually defines **six constituent types**, not four — though the framing may feel like four primary "reader" types with two that sit differently. Here's what the document says:

Type D — System Declarant holds institutional priorities and budgets and has a clear understanding of constraints, costs, current realities, and future possibilities. They see the destination and the potential a pathway holds.

Type C — Evaluator and Pathway Designer answers Type D's vision with a feasibility and viability response. They conduct analysis and produce the first level of the *how* — practical pathways or options for pathways — for the consideration of Type D and the system.

Type B — Constructor and Lead Designer leads the designing of the way, constructing and building the pathway in practice — translating what Type C has made feasible into something actually built and delivered.

Type A — Agent of Change is at the frontline, where the pathway meets the world. They activate the pathway and are the human interface between the system and the person the pathway serves.

Type P — The Beneficiary is the ultimate purpose of the pathway. Others walk the pathway too — designing it, building it, activating it — all in service of Type P. Type P is not an audience in the same sense as the others; they are the purpose. The pathway only has meaning if it works for Type P.

There is also a **Type 0**, which is the overall story weaving vision and feasibility for all audiences.

What Farmers Are Trying Now

Use Cases on the DPI + AI Pathway for Agriculture

OAN Diffusion Pathway — Package of Pathways | Type A

A Note on This Document

This is a record of what farmers are actually doing — the questions they are bringing to AI systems built on this pathway, and what those questions reveal about what they need.

It is not a list of features. It is not a design specification. It is an account of where farmers are meeting this technology in their lives, drawn from deployments that are live and in use.

Two contexts are covered: crop farming and dairy farming. The use cases differ in their particulars, but the underlying pattern is the same. Farmers are coming with practical, immediate, situated questions. And they are finding answers.

Crop Farming

Asking about their crops and what is going wrong

Farmers are bringing their fields to the system — describing what they are seeing, asking what it means, and asking what to do. A leaf that has changed colour. A pest that has appeared. A yield that is lower than expected. These are the questions that, without access to a system like this, would have had to wait for an extension worker to visit, or go unanswered. Farmers are finding that when they describe a symptom, they get a response that is grounded in verified agricultural knowledge — not a guess, not a generic suggestion, but an answer that draws from crop science and pest management research and tells them what the problem likely is and what steps to take.

This is the most frequently asked category on deployments that have walked this pathway.

Asking about the weather

Farmers are using the system to check the weather before they make decisions — whether to sow, whether to spray, whether to move produce. They are asking about what is coming in the next few days, and sometimes about what has already happened — the rainfall last week, the temperatures over the past month — **to make sense of what they are seeing in their fields.**

They are getting answers that are specific to where they are, drawn from meteorological sources, in a language they speak.

Asking about market prices

Farmers are asking what their crop is worth — at the nearest market, for the variety they have grown, at the time they are ready to sell. **Price information of this kind was previously hard to find, slow to travel, and often reached the farmer after the decision had already been made.**

Farmers are now asking the question and getting a current, sourced answer — one that is specific to their location and their crop — before they decide when and where to sell.

Asking about government schemes

Farmers are asking whether they are eligible for schemes they have heard about, what is required to apply, and what the benefit is. These are consequential questions — schemes for insurance, for income support, for subsidised inputs — and **the answers have often been difficult to obtain without navigating bureaucratic processes or finding someone who knows.**
Farmers are finding that they can ask in their own words and get clear, accurate information about what they qualify for and what the process involves.

Asking about the status of an application they have already made

Farmers who have applied for a scheme want to know what has happened to their application. They are asking the system and getting a real-time answer drawn from benefit registries — not a general description of processing timelines, **but the actual status of their own application.** Whether it has been approved. Whether a payment has been made. Whether something is outstanding.

Asking questions that draw on who they are and where they farm

Where a farmer's identity, land holding, and crop information are held in a registry, the system can answer questions in a way that is specific to them — not advice about a crop in general, **but advice about their crop, on their land, in their location, in this season.** Farmers are finding that the system knows enough about their situation to give answers that actually fit.

Asking where to find things they need

Farmers are asking about services and infrastructure near them — a machine to hire for harvesting, a facility to test their soil, a place to store their produce, a veterinary service for their animals. **This information exists, but it has not always been easy for a farmer to find.** The system draws from service registries and tells the farmer what is available and where.

Asking how to reach the people who are supposed to help them

Farmers are asking for the contact details of the agricultural officer assigned to their area, or the nearest agricultural knowledge centre. **These are human connections — and the system is not replacing them. It is making them findable.** A farmer who could not easily reach the person assigned to support them can now find out who that person is and how to get in touch.

What Farmers Are Beginning to Try

Beyond the questions that are now routine, farmers are beginning to try things that go further — **moving from asking to doing**.

Some are beginning to use the system to work through a scheme application, guided step by step through what is required, without having to visit a portal or a government office.

Some are beginning to explore what credit products might be available to them — using the same interface where they asked about their crop or their scheme.

Some are beginning to raise grievances through the system — when a scheme has not delivered, when an application is stuck — and to track what happens next.

And in some places, farmers are beginning to contribute as well as receive — describing what they are observing in their fields, and in doing so, adding to the stock of information that the system draws on.

These are early. But they are being tried.

Dairy Farming

Asking about an animal that does not seem well

Dairy farmers are bringing their observations to the system — a change in an animal's behaviour, a drop in milk, a symptom they cannot identify. **They are describing what they see and asking what it might mean.** The system draws from veterinary knowledge and the animal's own health records, and gives an early assessment and a suggested response. **For farmers who do not have immediate access to a veterinary professional[**Amul data of 1400 vets for 3.6 million farmers and 22 million cattle

], this can be the difference between catching a problem early and losing an animal.

Asking when their animals are due for vaccinations

Farmers are using the system **to keep track of** which animals need which vaccinations, and when. The system draws from individual animal records and sends reminders that are specific to each animal's history — not a general vaccination calendar, but a reminder for this animal, based on what has already been given and what is next due.

Asking what to feed their animals

Farmers are asking for feeding guidance that fits their animals' actual condition — their stage of lactation, their pregnancy status, their health history. The system draws from each animal's profile and the farmer's available inputs **and gives advice that is tailored**, not generic. **The goal is better milk yield and better animal health,** through feeding that is matched to what the animal actually needs.

Asking about breeding and reproductive cycles

Farmers are using the system to track when their animals are likely to be in oestrus and when insemination is due. Missing a cycle has a direct cost — in time, in productivity, in the animal's welfare. **Farmers are finding that the system, drawing from reproductive records, can keep track of what they would otherwise have to hold in memory or on paper, and remind them at the right moment.**

Asking why their milk yield has changed

When milk production drops, or fails to improve, farmers want to understand why. They are bringing the question to the system and getting a response that draws from their own production history — the pattern of their herd's output over time — and helps them **identify whether the cause is nutritional, health-related, or seasonal, and what to adjust**.

Asking about the treatment history of a specific animal

Farmers are asking what has happened to a particular animal — what conditions it has had, what it has been treated with, what vaccinations it has received. This is information that, in a well-resourced cooperative structure, is held in a database. **The system makes it accessible to the farmer in a conversational form, so they can recall it when they need it — before a veterinary visit, when making a decision about an animal, when something goes wrong.**

Asking what to feed when fodder is scarce or conditions have changed

Farmers are asking for feeding advice that accounts for what is available in their area — not just what is ideal in principle. The system draws from regional data on fodder availability and production, and gives suggestions that are grounded in what the farmer can actually find and use.

Asking about their own milk procurement and payment records

Dairy farmers want to understand their own relationship with the system they supply — how much milk they have delivered, what quality it has been graded at, what they have been paid, and what has been deducted. They are asking the system and getting answers drawn from their own transaction records, in their own language, through a voice call.

Asking about schemes and insurance available to them

Dairy farmers are asking about schemes specific to livestock — insurance for their animals, support for veterinary costs, programmes from the cooperative or from government. The system draws from scheme and programme registries and tells the farmer what is available, what they are likely to qualify for, and how to pursue it.

Asking questions they have always had, in a language they actually speak

For many dairy farmers — and in particular for the women who form the majority of cooperative dairy producers in some regions — a voice call in their own language is the only channel that works. They have never had a smartphone. They may not read with ease. **The expert knowledge that could help them manage their animals has existed, but has not been reachable**.

They are now reaching it. Through a call, in their language, at any hour. **The questions they are asking are the same questions they have always had. What has changed is that there is now a system that can answer them.**

What Is Being Learned

Across crop farming and dairy farming, what is being learned is simple: **when a system answers the real questions farmers are asking — in their language, through a channel they can use, with an answer that fits their situation — farmers use it**.

They are not being persuaded. They are asking questions that matter to them, and finding that the answers are there.

The pathway being walked is from information to action. The first questions — about a crop, about the weather, about a scheme — are opening into a wider set of things farmers are beginning to try: applying, tracking, resolving, contributing. Each step has only been possible because the previous one was walked.

Those who are considering walking this pathway will find that the early steps are already there. The question is simply: how far do you want to go?

What’s the farmer experience is

The pathway **reaches farmers through multiple channels**, designed around the reality of rural life: a mobile app for those with smartphones, a chatbot accessible via WhatsApp, a web interface, and a voice call line that works on any phone including a basic feature phone. In Maharashtra, the short code is 155313. In Bharat-VISTAAR, feature phone farmers can call 155261 in Hindi or English. In Gujarat, Amul's Sarlaben is reached at 08035453545. In each case, the entry point is a call or a message — not a registration form, not an app download requirement, not a prerequisite of digital fluency.

The farmer does not need to know what the system knows about them. When a farmer calls or types a question, the system — with their consent — draws from their registered Farmer ID, Farm ID, crop-sown records, and plot data to personalise the response. A farmer who has sown wheat in Shirpur does not get a generic wheat advisory. They get a response calibrated to their crop, their location, the current season, and the current weather forecast from IMD. The system already knows what they grow and where. The farmer just has to ask.

Two modes of interaction have emerged in practice. **The first is inbound** — the farmer comes with a question. In December 2025, MahaVistaar was receiving over 440,000 categorised queries a month. The largest single category was crop and pest advisory: 205,000 queries. Farmers asked "Leaf curling in chilli — what to do?" and "Can guava be grown on my land?" They asked for the weather in Parbhani tomorrow, for onion prices near Nashik, for the status of their drip-irrigation subsidy application, for the name of the agriculture officer assigned to their village. These are not abstract needs. They are the actual questions farmers have had, unanswered, for decades.

The second mode is outbound — the system comes to the farmer. A cotton farmer in Vidarbha, registered on the platform, receives a push notification when pink bollworm is detected in their region: "An infestation has been detected. Please inspect your crop immediately and take control measures." Before the next rains, they receive a voice call: "Heavy rainfall is expected in your area in the next 24 hours. Please take necessary precautions." Across the crop calendar — from sowing through harvest — the system delivers roughly 15 stage-based advisories proactively, without the farmer needing to ask. The shift from a platform that waits for questions to one that anticipates needs is what the second phase of MahaVistaar is designed to complete.

For the dairy producer in Gujarat, the interaction looks different but rests on the same foundation. A woman producer in one of Amul's 18,500+ villages calls Sarlaben with a question about her cattle. The system draws from Amul's cooperative data — 2 billion milk procurement transactions, records on 30 million cattle, integrated veterinary doctor records — to give her an answer that is specific to her herd, her animal, her circumstances. The name Sarlaben was chosen deliberately: to signal that this is a system built for her, in her language, on her terms.

2. The Problem This Pathway Addresses — As System Leaders Found It

The system leaders who walked this pathway before did so because a structural problem had become legible to them. It is worth naming that problem plainly, because the pathway only makes sense as a response to it. The problem has two layers. **The first is about fragmentation**. The second — deeper, and often overlooked — is about **inclusion**.

The fragmentation problem

A Marathi-speaking farmer in Raigad district asks: my paddy is still standing — should I harvest now? The answer requires current weather data, local market prices, and crop-stage advisory from the state agricultural university. Each of those sits in a different institution. **None of those institutions, individually, can answer the question. Together, they can. But they were never designed to work together.**

The state agricultural university holds the knowledge. The weather service holds the forecast. The market system holds the prices. The scheme department holds the eligibility rules. The grievance portal holds the application status. For the farmer, these might as well be in different countries. **There is no single window. There is no shared understanding of who the farmer is, what they grow, or what they need.** And even if there were, the farmer would need to navigate each system separately — logging in, reading, searching, interpreting — **to piece together an answer that the system, in aggregate, already holds.**

In Bihar, historically, barely one agriculture field officer served every 2,000 farmers. In Maharashtra, extension officers travelled long distances answering questions across crops, livestock, schemes, and markets, carrying scattered PDFs, circulars, and personal notes. This was not a failure of individuals. **It was a structural constraint: advisory capability that scales with population requires a different architecture than one that scales only with staff.**

The inclusion problem — why farmers cannot self-include

This is the part of the problem that digital systems have consistently underestimated.

Farmers are not excluded from institutional knowledge because the knowledge doesn't exist. **The knowledge exists — in abundance.** Agricultural universities publish research. Government departments maintain scheme portals. Market systems publish prices. Weather services publish forecasts. All of it is, technically, available. A determined person with a smartphone, reliable internet, sufficient literacy, enough time, and fluency in the right language could, in principle, piece it together.

That description **excludes most of the people the systems were built to serve**.

The smallholder farmer in rural Maharashtra or Oromia does not have reliable internet. Many do not have smartphones — they have basic feature phones. Many cannot read the language the portal is in, or cannot read at all. The websites that hold agricultural guidance were not designed for them: they require navigation, registration, login, scrolling, decoding of government terminology, and an ability to triangulate across multiple sources. Even farmers who try to use these systems — and many do try — encounter a cascade of barriers before they reach anything useful. They search. They land on the wrong page. They cannot read the PDF. The portal asks them to register. Registration requires documents. The advisory is in Hindi. They give up.

This is not occasional. **It is structural.** The people who most need the advisory are precisely the people who cannot access it through the channels designed to hold it. **Digital proliferation has produced an abundance of information that flows to those who are already connected, literate, resourced, and fluent — and stops at the edge of that circle.**

<u>The farmer is not failing to include themselves. The system is failing to reach them</u>.

What could not be accessed before - Voice as the structural solution to inclusion

voice is not a feature of the OAN pathway. **It is the architectural response to the inclusion problem.**

A farmer who cannot read does not need to read. A farmer who does not have a smartphone can call a number. A farmer whose primary language is Marathi, or Gujarati, or Oromo, does not need to translate. **Voice is the channel through which every other barrier dissolves:** literacy, device type, navigation complexity, language, and portal registration all become irrelevant. The farmer speaks their question in their language, on the phone they already own. **<u>The system listens, understands, assembles the answer from institutional sources, and speaks back.</u>**

Nandan Nilekani named it plainly: using a basic feature phone, a Marathi-speaking farmer can now consult a chatbot for advice on soil, seeds and irrigation, and receive guidance in their own language. When the monsoon is delayed, people want to know who stands behind a recommendation. **Citizens tend to trust institutions rather than algorithms.** For AI to be adopted by a whole population, it must carry the credibility of trusted institutions, with traceable and verifiable authority. Voice-first access built on verified institutional data is how both conditions — reach and trust — are met simultaneously.

The MahaVistaar short code is 155313. Bharat-VISTAAR is reachable at 155261. Sarlaben answers at 08035453545. Feature phones, landlines, voice calls on basic devices. These are not concessions to constraint. They are the primary design.

The woman dairy producer in one of Amul's 18,500+ Gujarat villages — predominantly rural, lower digital literacy, conducting transactions that may represent a substantial portion of her household income — reaches Sarlaben by calling a number in Gujarati. The system knows her. It draws from 2 billion milk procurement transactions and 50 years of cooperative data to give her an answer specific to her cattle, her history, and her situation. She did not need to navigate a portal. She did not need a smartphone. She needed a phone and a question.

That is what inclusion looks like when the infrastructure is built around the person rather than assuming the person can build around the infrastructure.

The transformation system leaders pursued was institutional before it was technical

The Government of Maharashtra was not trying to build a standalone app. **It was trying to create a state-level advisory capability embedded in the workflows of farmers, extension workers, and departments — one that could move from fragmented advisory channels to a common rail for trusted intelligence, make institutional knowledge usable in the field in voice and natural language, create a feedback loop so field usage improved policy over time, and scale without adding staff proportionally.**

That is what the pathway offered. That is what they found it could do.

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3. Who Has Walked This Pathway — Context, Capacity, and Broadly Observed Outcomes

Five contexts have walked this pathway as of March 2026. Each brought a different institutional starting point. Each walked it under different constraints, with different populations in mind, and at different speeds. **What they share is the core decision to address the structural problem — fragmentation and exclusion — through open, federated, voice-first digital public infrastructure.**

The outcomes noted below are the early signals each context has generated. They are presented as what has been observed, not as what is guaranteed.

MahaVistaar — Maharashtra, India *A state government building the first deployment, from scratch,* ***without a playbook***

The Department of Agriculture, Government of Maharashtra, walked this pathway first. **It had no prior architecture to draw on, no governance playbook to reuse, no failure mode library to consult. It built from commitment to deployment in nine months — a span that produced not just a working system, but every transferable asset the pathway now carries.**

The context: Maharashtra is India's second-largest state by GDP, with 152 lakh hectares of Kharif cropland and a deeply **fragmented** agricultural advisory landscape. The state had advisory platforms, university knowledge bases, scheme portals, and weather services — **all operating in isolation**, none connected.

The capacity in which Maharashtra walked it: as the pioneer. The Department of Agriculture **authorised data sharing across its own systems and from its agricultural universities. It stood behind the institutional authority the system cited on every farmer interaction.** It absorbed nine months of operational learning — dialect variation, API instability, trust-building with farmers — that no subsequent context had to rediscover.

Broadly observed outcomes as of early 2026: 342,000+ unique users. 1.67 million+ farmer questions answered. 791,000+ sessions, indicating repeat and sustained engagement — not one-off usage. 17 lakh farmers reached daily through proactive personalised voice alerts. 97%+ positive feedback rate, sustained at 98.5% in the most recent measurement period. The system spans weather, pest advisory, mandi prices, 40+ government schemes, 307 APMCs, 4 state agricultural universities, and 203 warehouses — all accessible through a voice call or message in Marathi.

Bharat-VISTAAR — National India, Ministry of Agriculture and Farmers Welfare *A national ministry establishing the common rail for the entire country*

Bharat-VISTAAR is India's national DPI for agriculture — what UPI is for payments, extended to agricultural information and services. It was announced in the Union Budget 2026-27 in early Feb 2026 with an allocation of Rs. 150 crore, formally launched by Agriculture Minister Shivraj Singh Chouhan in Jaipur on 17 February 2026, and championed by Prime Minister Modi at the India AI Impact Summit.

The context: India has 120 million farmers across dozens of states, languages, crops, and agricultural systems. No single state platform can reach them. **The national architecture is the common rail on which state and cooperative platforms connect as nodes — each retaining local specificity while drawing on national coherence.**

The capacity in which the Ministry walked it: as the national integrator. Bharat-VISTAAR does not replace what Maharashtra, Gujarat, or Bihar have built. It amplifies them. Maharashtra connects through MahaVistaar, Gujarat through Amul AI, Bihar through Bihar Krishi. **Each node retains its identity; the national layer provides the shared knowledge base, the AgriStack**

integration, the ICAR advisory corpus, and the PM scheme connectivity that no single state could build alone.

Broadly observed outcomes at launch: Phase-1 live with 10 major central schemes including PM-KISAN, PMFBY, Soil Health Card, and Kisan Credit Card. ICAR crop and livestock guidance integrated. Real-time IMD weather and NPSS pest alerts operational. Feature phone access via 155261 in Hindi and English. Early evidence from the Saagu Baagu pilot in Telangana — on which the design drew — showed a 21% increase in yield per acre and 9% reduction in pesticide use for cotton farmers. Full central scheme integration targeted by May 2026, with state-level onboarding ongoing.

Amul's Sarlaben — Gujarat, India *The world's largest dairy cooperative placing AI in the hands of 3.6 million women producers*

Amul walked this pathway as a private sector cooperative, not a government. Its context was different: the data infrastructure, the institutional relationships, and the member trust had all been built over 50 years of cooperative transactions. What Amul needed was not the institutional alignment — that existed. **What it needed was the AI layer, the open network protocols, and the pathway know-how to deploy in weeks rather than months**.

The context: Amul serves 3.6 million milk producers, predominantly women, across 18,500+ villages in Gujarat. It processes 3.5 crore litres of milk daily. Its members — the primary producers — had never had direct access to the knowledge held in Amul's own cooperative data: 2 billion milk procurement transactions, records on 30 million cattle, 1,200+ veterinary doctors' records, 50 years of cooperative history. **That data existed. The intelligence it could generate had never reached the farmer.**

The capacity in which Amul walked it: as the fastest adapter yet documented on the pathway. **Three weeks from commitment to deployment — because the architecture, governance frameworks, and deployment playbooks had already been built by Maharashtra and did not need to be rebuilt.** Amul **adapted them to its cooperative context, its data foundation, and its members**.

Broadly observed outcomes: 3.6 million farmers reached at launch. 1 million+ app downloads. Voice access in Gujarati for feature phone and landline users. **Personalised advisory based on each animal's individual history — not generic livestock guidance but advice specific to the cattle the farmer is standing next to.** Referenced by Prime Minister Modi at the India AI Impact Summit 2026. Planned expansion to 20 Indian languages via Bhashini, reaching 20,000+ villages across 20 states.

Bihar Krishi — Bihar, India *A state government building its own award-winning platform, then connecting to the national layer*

Bihar walked a different version of this pathway. Bihar Krishi was not built as an OAN deployment — it was built as an independent state initiative under Bihar's 4th Agriculture Roadmap, designed and delivered with MicroSave Consulting and Gates Foundation support, on the recognition that Bihar's structural gap was severe and required its own response.

The context: historically barely one agriculture field officer for every 2,000 farmers. A predominantly rural economy. **Deep exclusion from timely advisory, scheme access, and market information.** 25% of the farmer population are women.

The capacity in which Bihar walked it: as a state that built first on its own terms, then connected. Bihar Krishi was a rigorous, ground-up effort to unify fragmented state agricultural services into a coherent platform — 50+ schemes, 15,000+ extension workers trained, 38 districts covered. When Bharat-VISTAAR launched, Bihar Krishi became its Bihar node — connecting to the national knowledge base, AgriStack, and ICAR data through the national layer. **OAN amplified what Bihar had already built; it did not originate it**.

Broadly observed outcomes since May 2025: 850,000+ farmers registered. Coverage across all 38 districts. 38,000+ scheme applications submitted. 20–25% monthly engagement rate. 20 million farmers reached through digital outreach. ET DigiTech Award 2025 (Gold). SKOCH Award 2025 (Gold).

OpenAgriNet Ethiopia *A national transformation institute establishing the pathway's first deployment outside India, in three months*

Ethiopia walked this pathway as the first country outside India to do so. It walked it fast — three months, compared to nine for Maharashtra — precisely because the architecture, governance frameworks, language pipeline methodology, data connector approach, and failure mode library had already been built. The OAN delegation formally presented the Indian experience at a socialisation workshop in Addis Ababa. What Ethiopia needed to build was an adaptation, not a construction.

The context: agriculture at 35% of GDP and 60%+ of workforce, 15 million+ smallholder households, and **severely limited access to reliable digital advisory.** Ethiopia is also the host of COP32, and climate intelligence is integrated into the system design from the beginning. The Digital Agriculture Roadmap 2025–2032 provides the national strategic framework, with 22 prioritised use cases across six solution areas.

The capacity in which Ethiopia walked it: as a national transformation institute leading deployment across a country of extraordinary agricultural diversity — in language, crop system, climate zone, and connectivity profile. ATI aligned the Ministry of Agriculture, international development partners, and technology enablers around a single national architecture rather than fragmented parallel efforts.

Broadly observed outcomes and ambitions at launch: formally launched February 2026, three months from commitment. Integration with Fayda (Ethiopia's national digital ID). Voice-first access in local languages for farmers without smartphones or broadband. Thirty million farmers targeted, including 14 million women. An 8% income boost within five years as the stated ambition. Designed to position Ethiopia as a regional leader in digital agriculture across Sub-Saharan Africa.

What these five contexts have in common

Each context is different. The implementing organisation was a state department, a national ministry, a global cooperative, a state government, and a national transformation institute. The populations served ranged from Maharashtra's crop farmers to Gujarat's women dairy producers to Bihar's scheme-excluded smallholders to Ethiopia's climate-vulnerable Oromia farmers. The deployment times ranged from nine months to three weeks.

What they share is the recognition that the problem was structural, that voice was the access mechanism that inclusion required, and that an open federated architecture — where institutions remain the authority and the AI is the delivery layer — was the design that both served farmers and earned their trust.

The pathway that each of them walked is the same pathway. What changed, with each step, was how much of it had already been cleared.

The Basic Frame — DPI Powered by AI

Before describing what the pathway consists of, it is worth establishing a shared understanding of the frame within which this pathway operates. DPI powered by AI is a specific kind of approach. It is not simply an AI application built for agriculture. It is something structurally different — and for a system leader or evaluator, that distinction matters.

What Digital Public Infrastructure is

Digital Public Infrastructure is shared, foundational digital infrastructure — built as reusable building blocks for public benefit, operating on open standards, designed to be interoperable across institutions and contexts.

The analogy that travels furthest is physical infrastructure. Roads and railways do not tell vehicles where to go. They create the common foundation on which many vehicles can travel, efficiently, in many directions, without each needing to lay their own road. DPI is the digital equivalent: shared rails on which many services can run, rather than each service building its own track from scratch.

India's experience with non-agricultural DPI provides the most concrete illustration of what this means at scale. Aadhaar, India's national biometric identity platform, now covers more than a billion people and has enabled 164 billion digital authentications. UPI, the Unified Payments Interface, processes roughly 20 billion transactions monthly and is now the world's largest real-time payment system. Together, they show what shared digital rails produce: foundational capabilities that every service can draw on without rebuilding identity or payments from scratch, with the benefits distributed across every institution and every citizen who operates on those rails.

Agricultural DPI extends the same logic into the agriculture sector. Foundational layers — national identity, digital payments, data exchange infrastructure — are extended with sector-specific building blocks: farmer registries, farm registries, crop-sown registries, soil health data, market data, institutional advisory databases. **Each building block is not a product. It is infrastructure — modular, interoperable, designed to be combined in different configurations for different purposes.**

What AI adds to the frame

AI does not replace the DPI frame. It operates within it, as a layer above it.

The distinction is precise and important. Institutions — agricultural universities, state agriculture departments, cooperatives, national research institutes — remain the source of truth. They generate authoritative data. They certify advisories. They run schemes. They respond to real events. **That institutional authority does not transfer to the AI system. It stays where it belongs.**

What AI provides is the interface, coordination and delivery layer: the capability to receive a farmer's voice query in their language, interpret intent, reach across multiple institutional data sources simultaneously, assemble an answer, and return it in the farmer's language within seconds. AI scales what institutions already know. It does not substitute for what institutions know.

This separation is by design. The data fabric — farmer registries, advisory databases, scheme systems, weather APIs — remains independent of any particular AI model. **AI models change rapidly; institutional data systems must remain stable**. A state agriculture department does not need an AI team to participate in this infrastructure. It simply needs to expose a trusted advisory API. The AI layer above it uses that API conversationally — without changing what the department does, how it certifies knowledge, or who holds authority.

The result is a two-way system. It brings trusted data and advisories to farmers at the moment of decision. And it returns signals — anonymised, aggregated — back to institutions: which advisories are being accessed, which crops have insufficient guidance, which regions are showing stress patterns. **Institutions listening at population scale, continuously learning from real field interactions, is something that no previous system of agricultural extension could achieve.**

What this means for how the system is designed

Three design commitments follow from this frame, and understanding them is useful for both system leaders and evaluators considering this pathway.

The first is that the system is federated, not centralised. Data does not move into a single repository. Each institution retains ownership of its own data. The AI connects to data sources at the moment of query, with the farmer's consent, and returns results that cite their institutional origin. The farmer does not receive an AI opinion — they receive institutional knowledge, delivered through an AI interface.

The second is that voice-first is an architectural requirement, not a feature. Inclusion is built into the design from the foundation. A farmer who cannot read, who uses a basic phone, who lives with intermittent connectivity, reaches the system through a voice call. The system speaks back in their language. This is not achieved by adding voice as an afterthought to a text-first system. It requires the language infrastructure — multilingual models, local glossaries, voice-optimised pipelines — to be built as core components.

The third is that the AI layer is modular and the AI components are replaceable. No AI model is permanent. The infrastructure beneath it — the institutional data sources, the open network protocols, the farmer registries — remains stable as models evolve, are replaced, or are upgraded. **This means the system does not lock the institution into a single vendor or a single model. It preserves institutional continuity while enabling continuous AI improvement.**

Two examples of the frame in practice

MahaVistaar — the state agricultural system on shared rails

Maharashtra had all the components. The state had agricultural university knowledge bases. It had weather services from IMD. It had market data from 307 APMCs. It had 40+ government schemes in MahaDBT. It had farmer registries in AgriStack. None of these talked to each other. A farmer had no way to reach all of them from a single point.

MahaVistaar is the open network — built on OAN's open source building blocks — that places all of these on shared rails. The AI agentic orchestrator sits above those rails: it receives the farmer's voice query in Marathi, understands intent, reaches simultaneously to the university advisory database, the IMD weather API, and the APMC price feed, assembles the answer, cites its institutional sources, and speaks back to the farmer in Marathi. **The state agricultural university did not become an AI organisation. It continued certifying advisories.** The AI made those advisories reachable by a farmer in Kihim village on a basic phone.

Amul's Sarlaben — the cooperative data commons unlocked

Amul had something even more specific: fifty years of its own data. Two billion milk procurement transactions. Records on thirty million cattle, each with a unique ID tracking feed, health, treatment, and milking history. 1,200+ veterinary doctors contributing records annually. This data existed and had never spoken back to the farmer who generated it.

Sarlaben is the AI layer placed on top of that cooperative data infrastructure. When a woman dairy producer in Gujarat calls with a question about her cattle, the system draws from her specific animal's records — not from a generic livestock database, but from the actual history of that animal — and returns personalised guidance. The cooperative's institutional data remains in the cooperative's systems. **The AI reaches it, assembles the answer, and delivers it in Gujarati through a voice call to a feature phone. The institution did not change. The farmer's access to what the institution knew did.**

These two examples illustrate the same underlying frame at different scales and in different institutional contexts. The frame is not theory. It is what has been built, and what has been found to work.

What the Pathway Consists Of

The OAN pathway consists of two structural elements that only work together.

Open Source Building Blocks — the reusable technical components: the open network protocols, the agentic AI architecture, the voice-first access layer, the consent frameworks, the data integration standards. These are designed to be adapted to a new context, not rebuilt from scratch in each one.

A Collaboration Blueprint — the governance model, the enabler network, the institutional arrangements, and the documented organisational learning that make deployment possible and sustainable. This is the knowledge of how to bring institutions together: how to establish data-sharing agreements, how to authorise an AI layer to speak on behalf of public institutions, and how to sustain delivery beyond a pilot cycle.

Neither element works without the other. Building blocks without the collaboration blueprint produce technical deployments without institutional trust. A collaboration blueprint without the building blocks produces coordination without delivery.

What the pathway provides is both — and the accumulated knowledge of how others have assembled them in real contexts, against real constraints.

What Walking It Has Required — The System Leadership View

A commitment to the institution as the authority, not the technology. *Every deployment in this pathway is built on a single design principle:* ***the AI does not become the advisory authority. The institution does.*** *MahaVistaar cites the Department of Agriculture, the state agricultural university, IMD. Sarlaben cites Amul's cooperative knowledge base and veterinary records. The farmer trusts the system because the institution stands behind every answer. For system leaders, this means the decision to walk the pathway is also a decision* ***to authorise your institution's knowledge to be made accessible — and to stand behind what it says.***

A decision to align institutional priorities, not just commission a product. *In Maharashtra, the Department of Agriculture did not simply commission a platform.* ***It aligned data-sharing across universities, weather services, market data systems, and scheme delivery platforms. It authorised that data to flow through a common open network. It established the governance for what the AI could and could not do. That alignment was the system leadership work*** *— and it was the work that made everything else possible.*

A nomination of institutional ownership, named and accountable. *The pathway evidence from Maharashtra shows the first concrete decision is: nominate an Agri Secretary sponsor and nodal officers across agriculture, IT, and field operations.* ***A pathway cannot emerge from vendor activity alone.*** *It needs named public ownership. That is a system leadership decision.*

A willingness to begin bounded. *System leaders who have walked this pathway found that* ***starting with bounded ambition — one or two crops, a few districts, one or two languages — produced the early proof that enabled scaling****. MahaVistaar began with Kharif crops in selected districts. The national platform, Bharat-VISTAAR, was built on the architecture Maharashtra established. The Ethiopian deployment drew on both. The pathway favours leaders who can commit to a real start rather than wait for a perfect one.*

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5. What Walking It Has Required — The Pathfinder

Mapping what already exists before specifying what to build. *In every deployment, the first evaluator task was not specification —* ***it was inventory.*** *Maharashtra had fragmented advisory platforms, state agricultural university databases, weather services, and market data that each* *worked in isolation.* ***The feasibility question was: what can be joined, and under what governance? The answer shaped everything that followed.***

Assessing the institutional data question directly. *The OAN approach is built on the principle that* ***data stays with its legitimate owner. The AI does not copy data into a central repository — it connects to data through APIs, with farmer consent, at the moment of query.*** *For the evaluator, this means the feasibility assessment must address: which institutions hold what data, under what governance can it be accessed, and what consent architecture is required. In Bihar, this was the core design work — and the results, rigorously built, produced a platform that reached 850,000+ farmers across all 38 districts, onboarded 50+ government schemes, and earned two national awards.*

Understanding what the pathway already provides. *One of the most consequential feasibility findings for any new context is this:* ***the architecture, governance frameworks, language pipeline methodology, data connector approach, model evaluation benchmarks, and failure mode library from existing deployments are available.*** *Ethiopia's three-month deployment was possible precisely because its evaluators could assess what had already been built and determine what needed adaptation rather than construction.* ***The evaluator question shifts from "can this be built?" to "what does adapting this require here?"***

Evaluating sustainability from the beginning. *The OAN architectural principles establish that DPI must be sustainably affordable at national scale —* ***designed for declining cost*** *and minimal footprint. The evaluator working in any context must address: what are the ongoing operational costs, who bears them, and what incentive structures sustain participation over time? This is not a question the pathway answers prescriptively.* ***It is a question each context must work through, with the benefit of knowing what others have found.***

Treating the pilot as a learning phase, not a launch event. *The pathway evidence points consistently to one design: an eight-week pilot followed by structured review, before any decision on scaling.* ***What matters is not the first deployment by itself — it is the know-how it produces.*** *Evaluators who treat the pilot as the learning phase it is find that the pathway compounds. Those who treat it as a vanity launch find the compounding stops there.*

Examples of systems brought together from mahavistaar, AMUL, BV all supported here.

OpenAgriNet does not build what already exists. It connects it.

The 54 enablers listed here were not assembled for MahaVistaar. Most of them predate it — government schemes, university publications, weather services, language models, institutional mandates. They existed in separate systems, serving separate purposes, <u>with no shared route to the farmer</u>.

What OAN contributed was a common network layer — one that allowed these systems to be discovered, combined, and delivered as a coherent response to a single query from a field.

Systems organised here across four layers:

A. Institutional and governance — the funders, orchestrators, and government bodies that make the system legitimate and sustainable.

B. Technology and AI — the research institutions, language models, and data contributors that make it functional.

C. Structured data — the live feeds and system-generated datasets that keep it current and contextual.

D. Knowledge and documents — the accumulated guidelines, publications, and practice manuals that make its responses trustworthy.

This list is therefore not a catalogue of what was built. It is a record of what was brought together.

A. Institutional & Governance Enablers

Funders & Support

1. Gates Foundation
2. World Bank
3. UNDP
4. EkStep Foundation

Orchestration & Implementation 5. EkStep 6. COSS 7. Samagra 8. Artha Global

Government Bodies 9. Department of Agriculture, Maharashtra 10. Department of Livestock (MH) 11. Department of Fisheries (MH) 12. GoI Agri Ministry 13. DC Nandurbar 14. Maharashtra DBT

B. Technology & AI Enablers

Research & Development 15. IIT Mumbai 16. IISc 17. Vassar Labs 18. India AI Mission

Language & AI Models 19. Bhashini 20. AI4Bharat 21. Karya

Knowledge Contributors 22. ICAR Universities / Market Info

C. Data Enablers (Structured / System-Generated)

23. APMC Price Data
24. Warehouse Data (WDRA)
25. Custom Hiring Centre (CHC)
26. KVK (Krishi Vigyan Kendra)
27. IMD Forecast Data
28. Skymet Weather Data
29. Officer Data
30. Scheme Details Data

D. Knowledge & Document Enablers

State & National Programme Guidelines 31. Nanaji Deshmukh Krishi Sanjivani Prakalp (Phase 2) 32. Digital Agriculture Mission Operational Guidelines 2024 33. State Farmer Registry Handbook 34. PMFBY Guidelines 35. PMKSY Guidelines 36. ATMA Guidelines 37. PDMC Guidelines 38. NMEO-OS Guidelines 39. Agmarknet 40. DBT in Agriculture Guidelines

Crop Knowledge 41. Crop PoPs — MPKV, VNMAU, PDKV (Wheat, Rice, Sugarcane, Pulses, Oilseeds, Onion, Garlic) 42. MPKV Publications 43. Jaivik Kheti Guidelines

Livestock & Animal Husbandry 44. Standard Veterinary Treatment Manuals 45. Animal Health & Nutrition Modules (MANAGE, NDDB, GoM series) 46. Dairy, Poultry, Goat, Pig Husbandry Training Modules 47. Livestock Census Instruction Manual 48. Veterinary Diagnostics & Disease Manuals

Fisheries & Aquaculture 49. Fisheries & Aquaculture Guidelines (NFDB, RAS, Cage Culture, Seaweed Farming) 50. ICAR Technologies (Biopesticides, Biofertilizers, High-Value Seaweeds, Fish Breeding) 51. NBAIM Technologies Compilation

Research & Reports 52. Annual Reports — ICAR, DAFW 53. University Publications — MPKV, VNMAU, PDKV, GAVASU 54. National Programme Guidelines — NPOP, Bamboo Mission, Aatmanirbhar Pulses

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Total: 54 enablers across four layers.

Here is a comprehensive enumeration of MahaVistaar's system components as depicted in the architecture diagram and described across the narrative, including the hardware considerations that must be factored in.

This represents a replica across all of OAN instances to large degree.

It can be seen in 7 layers-

Layer 1 — User Layer The three classes of actors the system serves: Farmers (crops, livestock, fisheries), the Extension System (field officers and advisors), and Government & Policy (state planning, policy makers, ecosystem actors). **This layer defines who the system is accountable to and what kinds of decisions it must support.**

Layer 2 — Interface Layer (Natural Language Agricultural Interface) The channels through which users access the system: Push Notifications, Push Telephony, Chat/App, and Phone. Supports Marathi, Hindi, Bhili, and English. **This layer abstracts all interaction modality differences — text and voice both converge downstream into the same processing pipeline.**

Layer 3 — Moderation Layer (Safety, Security & Policy Enforcement) The first gate every query must pass. An independent model — fully decoupled from the advisory engine — that performs domain validation, content safety filtering, prompt injection defense, and input sanitization. **Classifies queries as Valid Agricultural, Non-Agricultural, Unsafe/Harmful, or Policy-Sensitive.** In voice, this is embedded within the ASR pipeline rather than a separate call.

Layer 4 — AI Decision Engine (Reasoning, Orchestration & Response) The core intelligence layer, itself composed of three tightly coupled stages: **the Reasoning Layer** identifies intent and structures the decision task; **Tool Orchestration** breaks it into tool calls, executes them in sequence, and enforces prescribed flows; and **Response Generation** synthesizes results into a grounded, source-attributed answer in the farmer's language. Nothing is answered from memory — every claim flows through a verified tool call.

Layer 5 — Knowledge & Scientific Models Domain-specific advisory knowledge and analytical models accessed by the orchestration layer: Crop Advisory, Livestock Health, Fisheries Models, and Weather & Market Analytics. Sourced from ICAR, universities, and research institutes. Also houses the bilingual Marathi↔English glossary that bridges the language gap without requiring a full corpus translation.

Layer 6 — Live Data & Institutional Sources The real-time and reference data feeds that ground every factual claim: IMD & Skymet (weather), AgriStack & Land Records (farmer identity and farm data), MahaDBT & Schemes (government subsidies and scheme status), Soil Labs (soil testing), and KVKs & KCC (local extension services). These are accessed as tools, not pre-loaded into the model.

Layer 7 — Digital Public Infrastructure (DPI) The interoperability and data exchange foundation that connects everything: OpenAgriNet, VISTAAR DPI, the Government Ecosystem, and the Data Exchange layer. Uses open protocols like Beckn to enable standardized discovery and interaction across independent institutional providers. This is what makes the system a piece of public infrastructure rather than a standalone application — it is designed to be plugged into and built upon.

Further detailing these layers.

MahaVistaar — System Component Enumeration

Layer 1: User-Facing Interface (Natural Language Agricultural Interface)

This is the outermost layer through which farmers, extension officers, and government users interact with the system.

Channels:

- **Push Notification** — proactive alerts sent to farmer devices
- **Push Telephony / IVR** — outbound and inbound voice calls over telephony networks
- **Chat / App** — text-based conversational interface (likely WhatsApp or a dedicated app)
- **Phone (voice)** — real-time voice interaction

Users served: Farmers (crops, livestock, fisheries), Extension System (field officers, advisors), Government & Policy (state planning, ecosystem actors)

Hardware considerations:

- Smartphones and feature phones used by farmers — the system must work across low-end Android devices and basic USSD/IVR phones
- Network edge infrastructure — low-bandwidth and intermittent connectivity in rural Maharashtra means the interface must be lightweight and tolerant of packet loss
- Telephony gateway hardware (SIP servers, PSTN interconnect equipment) for voice channel delivery

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Layer 2: MahaVistaar AI Decision Engine

This is the core processing brain, composed of four sub-components:

A. Moderation Layer (Safety, Security, Policy Enforcement)

- Independent language model (SLM or LLM) separate from the main advisory model
- Performs domain validation, content filtering, prompt injection defense, and input sanitization
- In voice channels, moderation is embedded within the ASR pipeline rather than as a standalone call
- Maintains adversarial test sets of up to 500 attack patterns

B. Reasoning Layer (Intent)

- Interprets farmer queries in context — what they need, what data is required, what sequence of retrievals will help
- Transforms natural language into a structured decision task
- Maintains session context for continuity across follow-up queries

C. Tool Orchestration

- Functions as an agentic executor — breaks intent into sub-tasks, selects tools, executes API call sequences, and synthesizes results
- Enforces prescribed query flows: Pest → Glossary → Advisory; Weather → Location → Forecast; Market → Location → Mandi Prices
- Guardrails: no tool = no answer; no source = no claim

D. Response Generation

- Constructs the final response in local language (Marathi, Hindi, Bhili, English)
- Every claim is grounded in verified tool output with source attribution
- Text-to-Speech component for voice output

Hardware considerations for the Decision Engine:

- **GPU servers** — LLM inference (both the moderation model and main advisory model) requires GPU acceleration; the document explicitly discusses GPU utilization as the primary cost driver when self-hosting open-source models
- The team is planning a transition from commercial APIs (per-token cost) to self-hosted open-source models, which fundamentally shifts infrastructure to owned or cloud-rented GPU compute
- GPU utilization rate directly determines cost per conversation — full utilization = low cost, underutilization = expensive
- Spare GPU capacity can be used for evaluation runs, synthetic data generation, and telemetry analysis

- **Low-latency inference hardware** is especially critical for voice: the model must begin streaming output before the full response is ready, so that TTS can start speaking immediately

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Layer 3: Voice Pipeline (ASR + TTS + Turn Detection)

Distinct from the decision engine, the voice channel has its own hardware-sensitive pipeline:

- **ASR (Automatic Speech Recognition)** — converts farmer speech to text; must handle Marathi, Hindi, Bhili, and regional accents
- **Turn Detection** — determines when the farmer has finished speaking
- **TTS (Text-to-Speech)** — converts advisory text back to natural-sounding speech in local language
- **Lightweight orchestration layer** — manages switching between ASR/TTS model configurations

Hardware considerations:

- Real-time ASR and TTS are latency-sensitive; dedicated inference endpoints (possibly separate from the main LLM servers) are needed
- Telephony interface hardware / cloud telephony (Exotel, Tata Tele, etc.) with low jitter and stable audio codec support
- The future direction is unified speech-to-speech models that handle the entire pipeline end-to-end — this will require newer, larger GPU hardware with higher memory bandwidth

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Layer 4: Knowledge & Scientific Models

Domain-specific models and knowledge bases accessed by the Tool Orchestration layer:

- **Crop Advisory Models** — crop SOPs, pest guides sourced from ICAR and universities
- **Livestock Health Models** — veterinary advisory knowledge
- **Fisheries Models** — fisheries-specific advisory content
- **Weather & Market Analytics** — real-time data aggregation and analysis models
- **Multilingual Glossary** — bidirectional Marathi↔English agricultural terminology bridge (critical language component)
- **Advisory Corpus** — chunked, metadata-tagged knowledge documents organized by crop → pest → treatment → source

Hardware considerations:

- **Vector database infrastructure** — for semantic search and RAG (Retrieval-Augmented Generation) over the advisory corpus; requires fast SSD-backed storage and sufficient RAM for index serving
- **Embedding model compute** — generating and querying vector embeddings at query time requires CPU or lightweight GPU nodes
- Storage sizing for advisory corpus documents — while content is relatively static, structured chunked storage at scale requires careful capacity planning

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Layer 5: External Data Sources & Integrations (Tool Ecosystem)

These are the live data feeds that ground every response:

- **IMD & Skymet** — real-time weather forecasts and historical data
- **AgriStack & Land Records** — farmer identity, farm registry, crop sown registry
- **MahaDBT & Government Schemes** — scheme eligibility and status (subsidy discovery, grievance, tracking)
- **Soil Labs** — soil testing data
- **KVKs (Krishi Vigyan Kendras) & KCC (Kisan Call Centre)** — local extension services and officers
- **APMC (Agricultural Produce Market Committee)** — mandi/market price data

Hardware considerations:

- API gateway infrastructure to handle concurrent outbound calls to multiple external services, with timeout and fallback logic
- Some data sources (especially weather stations) have geographic gaps — the system has expanded search radii and built graceful fallbacks, which requires intelligent routing logic
- Caching layer (Redis or equivalent) for frequently accessed data like weather forecasts and mandi prices, reducing repeated API calls and improving latency; requires in-memory hardware capacity

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Layer 6: Digital Public Infrastructure (DPI) Foundation

The lowest layer — the interoperability backbone:

- **OpenAgriNet** — open network protocol layer
- **VISTAAR DPI** — ministry-level data interchange
- **Government Ecosystem integrations** — connecting to state and central government data services
- **Data Exchange** — standardized discovery and interaction via Beckn protocol

Hardware considerations:

- Network infrastructure for secure government data exchange — dedicated leased lines or VPN tunnels may be required for sensitive data (land records, scheme eligibility)
- Compliance with NIC (National Informatics Centre) hosting requirements if government data residency rules apply
- Load balancers and API management gateways to handle burst traffic, particularly when large farmer groups or extension campaigns generate spikes

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Cross-Cutting Hardware & Infrastructure Concerns

Beyond individual layers, several system-wide hardware factors must be planned for:

Compute & Scaling:

- GPU cluster sizing must account for peak concurrent voice calls (real-time, latency-sensitive) versus batch workloads (evaluation, fine-tuning, synthetic data generation)
- Fine-tuned smaller open-source models (which the team has shown can outperform larger commercial APIs — 94% vs 91% on their eval framework) will require on-premise or cloud GPU for both training runs and serving

Security Hardware:

- HSMs (Hardware Security Modules) or equivalent for managing token-based authentication keys with time-bound access
- Rate-limiting infrastructure (IP-based throttling, adaptive rate limiting) to defend against "Denial of Wallet" attacks that target GPU/token budgets rather than network availability

Telemetry & Monitoring:

- Logging infrastructure for conversation telemetry — the system plans to use a small specialized model to scan conversations and flag issues; this model needs its own compute allocation
- Observability stack (metrics, traces, logs) requires storage and compute separate from the inference path

Resilience:

- Redundant connectivity for rural telephony gateways
- Fallback configurations for external data sources that have known gaps (e.g., weather station coverage)

In summary, MahaVistaar has six logical layers of software components, but the hardware story centers on three critical axes: **GPU compute for LLM inference** (the dominant cost and performance variable), **low-latency telephony and ASR/TTS infrastructure** (the dominant constraint for voice quality), and **fast, reliable data connectivity** to multiple government and real-time data sources. **The shift to self-hosted open-source models makes GPU capacity planning the most consequential infrastructure decision ahead.**


Q 3: What is the mandi price for wheat?
QUERY ON MARKET
A: You have sown wheat in Shirpur. The latest wheat prices there are between ₹2,350 and ₹2,620 per quintal from the most recent price update 3 days ago.
महाविस्तार - AI
Source: Agristack, Mandi Prices
महाविस्तार - AI
मराठी
विचारा
डुक्कर मध्ये मस्टायटिस (कासदाह) साठी काय उपचार आहेत?
जांभूळ साठी आदर्श सिंचन वेळापत्रक काय आहे?
आज एपीएमसी मुंबई मध्ये तांदूळ चा भाव काय आहे?
तुमचा प्रश्न विचारा...


Q 2: What is the status of my scheme application?
QUERY ON SCHEME
Your PMFBY policy is approved for the 2025-Kharif season, covering maize with a sum insured of ₹5.08 k.
Do you to know more?
महाविस्तार - AI
Source: AgriStack, MahaDBT Application Status
महाविस्तार - AI
मराठी
विचारा
मटकी मध्ये तण नियंत्रण पद्धती कोणत्या आहेत?
माझ्या भागातील पुढील ५ दिवसांचा हवामान अंदाज काय आहे?
कोल्हापूर मध्ये दोडका चा सर्वसाधारण भाव काय आहे?
तुमचा प्रश्न विचारा...
AI-निर्मित माहिती. कृपया स्वतंत्रपणे सत्यापित करा.


Q 1: What irrigation practice should I follow?
QUERY ON CROP
A: You have sown HD 2189 wheat in Akola (Vidarbha region) on 1 hectare of medium to deep black soil, using limited borewell irrigation. Here's your irrigation advice:
💧 If you can give one irrigation, do it around 40–42 days after sowing
💧💧 For two irrigations, apply the first at 20–22 days and the second at 60–65 days.
महाविस्तार - AI
Source: AgriStack, Dr. Panjabrao Deshmukh Krishi Vidyapeeth (Akola) & MPKV Rahuri Recommendations
महाविस्तार - AI
मराठी
विचारा
आज कोल्हापूर मध्ये शेंगदाणा चा भाव काय आहे?
अननस साठी आदर्श सिंचन वेळापत्रक काय आहे?
शेती यंत्रांसाठी एटीएमए योजनेत अनुदान कसे मिळेल?
तुमचा प्रश्न विचारा...
AI-निर्मित माहिती. कृपया स्वतंत्रपणे सत्यापित करा.